

| 院教学指导委员会主任<br>(院长) | 学院分管教学<br>(副院长) | 审核人<br>(专业责任教授团队负责人) | 执笔人 |
|--------------------|-----------------|----------------------|-----|
| 刘涛                 | 黄琼冰             | 钱功明                  | 叶青  |

## 矿物加工工程专业“卓越工程师”培养方案

### Curriculum for Undergraduate of Mineral Processing Engineering Major “Excellent Engineer Training Program”

#### 一、专业介绍

矿物加工工程专业始于1973年的武汉钢铁学院，已通过中国工程教育认证，是国家级一流本科建设专业点、国家“卓越工程师教育培养计划”、国家级特色专业、湖北省专业综合改革试点专业。本专业经过多年的发展与积淀，形成了领域特色鲜明、在国内外具有较大影响的创新团队，拥有“国家环境保护矿冶资源利用与污染控制重点实验室”、“战略钒资源利用省部共建协同创新中心”、“冶金矿产资源高效利用与造块湖北省重点实验室”等教学科研平台及多个国家级工程实践教育中心和大学生创新实践教学基地，为行业输送了大批优秀人才。

#### I. Professional Introduction

The Mineral Processing Engineering major began in Wuhan Iron and Steel College in 1973 and has passed the China Engineering Education Certification. It is a national first-class undergraduate construction major, a national "Excellent Engineer Education and Training Plan", a national characteristic major, and a pilot major for comprehensive professional reform in Hubei Province. After years of development and accumulation, this major has formed an innovative team with distinctive characteristics in the field and significant influence both domestically and internationally. It has teaching and research platforms such as the "National Key Laboratory for Environmental Protection, Mining and Metallurgical Resource Utilization and Pollution Control", the "Collaborative Innovation Center for Strategic Vanadium Resource Utilization between the Province and the Ministry", the "Hubei Province Key Laboratory for Efficient Utilization and Blocking of Metallurgical Mineral Resources", as well as multiple national engineering practice education centers and university student innovation practice teaching bases, providing a large number of outstanding talents for the industry.

#### 二、培养目标

本着“面向工业界、面向未来、面向世界”的工程教育理念，本专业立足矿物加工、资源综合利用、钢铁冶金和环境工程等相关领域，培养德、智、体、美、劳全面发展，社会责任感强，具有良好的人文和科学素养、合理的知识结构、厚实的基础理论知识，具备从事生产、工艺和设备设计、研究与开发、技术管理和经营的基本能力，工程实践能力强，具有一定的国际视野，能解决复杂工程问题的高素质创新型人才。期待本专业毕业生五年左右达到以下目标：

1. 具有良好的人文社会科学素养及较强的适应社会和环境的能力，具有良好的语言表达及沟通协调能力、团队意识和合作精神；

2. 具有社会责任感, 坚守工程职业道德, 具有与工作相关的质量意识、环境保护意识和安全意识;
3. 能熟练运用矿物加工工程及相关领域科学研究、工程设计和技术服务等工作所需的基础理论知识和现代技术手段研究复杂工程问题;
4. 在工作中能利用矿物加工过程的基础理论和工程实践知识分析、解决实际工程问题, 在新技术与新产品研发、工程设计和生产组织管理方面具备突出能力;
5. 具有较强的信息获取、理解能力, 能及时了解本专业相关学科前沿及发展动态, 具有终身学习的能力。

## **II. Training objectives**

Based on the engineering education philosophy of facing industry, facing future, and facing world, aiming at preparing all-rounded, a strong sense of social responsibility, good humanistic quality and international vision talents with certain innovation ability, team management and coordination ability, lifelong learning ability, this program is designed to provide a thorough grounding specialized knowledge and rich engineering practice ability in the fields of mineral processing, comprehensive utilization of resources, metallurgical and environmental engineering. Students with engineering science basic theory and specialized knowledge can solve complex engineering problems and qualified for production, design, research and development and management. Graduates of this major are supposed to achieve the following aims in about 5 years:

1. Good human and social science literacy and environmental and social adjustment and adaptability, good language expression and communication skills, team spirit and cooperation spirit. 1. Being in physical and mental health, possessing good humanity and social science literacy, having a strong ability to adapt to the society and environment, good language expression, communication skills, and team and cooperation spirit.
2. A sense of social responsibility; adhere to the professional ethics of the project, with quality awareness, environmental awareness and safety awareness.
3. Be familiar with the basic theory of mineral processing engineering and related fields, such as scientific research, engineering design and technical service, and modern technology method to study complicated engineering problem.
4. An ability to use the specialized knowledge of the mineral processing process solving the practical engineering problems and an excellent ability in the production of the new technology and new product development, engineering design and production organization management.
5. An ability to acquire information, understand the forefront and development trends of mineral processing and lifelong learning.

## **三、专业特色**

矿物加工工程专业面向国家发展战略需求, 服务矿业及区域经济社会发展, 经过多年的发展形成了矿产资源高效分选与提取、烧结与球团、资源综合利用、矿物材料等具有鲜明特色的研究平台。在微细粒矿物分选、矿物湿法冶金、高效磨矿理论与装备、烧结球团、资源综合利用等方面开展了创新性工作, 建成了“矿产资源物性特征-分离提纯-高值利用”的高素质专业技术人才培养体系, 在高效洁净矿物加工理论技术, 烧结球团与二次资源综合利用等方面特色鲜明。

## **III. Professional characteristics**

The Mineral Processing Engineering major caters to the needs of national development strategies, serving the mining industry and regional economic and social development. After years of development, it has formed a research platform with distinct characteristics, including efficient sorting and extraction of mineral resources,

block making and metallurgy, comprehensive utilization of resources, and mineral materials. Innovative work has been carried out in the fields of fine-grained mineral sorting, mineral hydrometallurgy, efficient grinding theory and equipment, iron ore block making, and comprehensive resource utilization. A high-quality professional and technical talent training system has been established for "mineral resource physical characteristics separation purification high-value utilization". The system has distinctive features in efficient and clean mineral processing theory and technology, as well as iron ore block making and comprehensive utilization of secondary resources.

#### 四、毕业要求

1. 具有良好的人文社会科学素养和高度的社会责任感、理解并遵守工程职业道德。
2. 掌握数学、自然科学、工程基础知识和技术经济管理知识,并能用于解决矿物加工复杂工程问题。
3. 掌握矿物加工工程的基本理论及专业知识并能用于描述和分析矿物加工复杂工程问题,以获得有效结论。
4. 具有对矿物加工工程及相关复杂工程问题进行实验、设计、研究和建模等工程综合知识和突出的工程实践能力,能够解决矿物加工复杂工程问题,具有创新意识。
5. 具备熟练使用信息获取工具和现代工程工具的能力,能够对矿物加工工程设计和研究中的复杂问题进行模拟和预测。
6. 能够在解决矿物加工工程相关问题时考虑生产、设计、研究与开发、安全和环境保护等方面的国家宏观发展相关产业政策与法律法规,正确认识和评价工程实践对环境、社会可持续发展的影响。
7. 具有开放包容的心态,积极沟通与分享,具有团队合作精神、组织管理能力。具有较强的书面和语言表达能力。
8. 了解学科前沿发展趋势,关注本专业与其他学科衍生交叉的新理论、新方法和新技术,能熟练运用一门以上的外语,具备一定的国际视野和跨文化的交流、竞争与合作能力。
9. 具有自主学习和终生学习的意识,能主动并持续更新核心知识以适应专业或职业发展。

#### IV. Requirements

1. Have a good social literacy and social responsibility, understand and abide by the professional ethics of the project.
2. Master mathematics, natural science, engineering knowledge, economic management knowledge and specialized knowledge, and can be used to solve the complex problems of mineral processing.
3. Have an ability to describe and analyze the complex engineering problems of mineral processing using theory and professional knowledge of mineral processing engineering.
4. Have a sense of innovation, the engineering knowledge and excellent engineering practice ability of mineral processing to experiment, design, research and model for solving complex engineering problems in mineral processing.
5. Have ability to use information acquisition tools and modern engineering tools and apply it to model and predict in the design and research of mineral processing engineering.
6. Understand the national industrial policies and laws related design, production, research and explore, safety and environmental protection in mineral processing, have ability to understand and evaluate the effect of engineering practice to the development of environment and society.
7. Have an open and inclusive attitude, ability of organizing and managing, expressing and communicating.
8. Understand the forefront of mineral processing, pay attention to the new theory, new method and new technology of the professional cross and other subjects derived, proficiency in the use of more than one foreign

language as well as international vision and ability to compete and cooperate in cross-cultural communication.

9. Have a sense of self-study and lifelong learning to update the core knowledge for adapt to professional development.

附：培养目标实现矩阵

|        | 培养目标 1 | 培养目标 2 | 培养目标 3 | 培养目标 4 | 培养目标 5 |
|--------|--------|--------|--------|--------|--------|
| 毕业要求 1 | √      | √      |        |        |        |
| 毕业要求 2 |        |        | √      | √      |        |
| 毕业要求 3 |        |        | √      | √      |        |
| 毕业要求 4 |        |        | √      | √      |        |
| 毕业要求 5 |        |        | √      | √      | √      |
| 毕业要求 6 |        | √      |        |        | √      |
| 毕业要求 7 | √      |        |        |        |        |
| 毕业要求 8 | √      |        |        |        | √      |
| 毕业要求 9 | √      |        |        |        | √      |

### 五、专业主干课程

流体力学、分析化学、物理化学、冶金物理化学、固体物料分选学、造块学、矿物加工研究方法、矿物分选综合实验、造块工艺学系统实验、矿物加工工程设计、科技创新、工程实践与创新等。

### V. Core courses

Fluid Mechanics, Analytical Chemistry, Physical Chemistry, Metallurgical Physical Chemistry, Solid Materials Separation, Agglomeration, Research Techniques of Mineral Processing, Systematic Experiments of Mineral Processing, Systematic Experiments of Agglomeration Technology, Design of Mineral Processing, Technology Innovation, and Engineering training and innovations.

### 六、基本学制：四年

### VI. Recommended length of the program: 4 years

### 七、授予学位：工学学士

### VII. Degree: Bachelor of Engineering

学生修满所规定的最低毕业学分，符合武汉科技大学授予学士学位规定，授予工学学士学位。

### 八、毕业学分要求：160学分

| 课程类型     |    | 学分要求 | 课程类型     | 学分要求 |      |
|----------|----|------|----------|------|------|
| 1、公共课程平台 |    | 43   | 3、专业课程模块 | 38   |      |
| 公共基础课程   |    | 28   | 专业必修课程   | 27.5 |      |
| 通识教育课程   | 必修 | 9    | 专业选修课程   | 10.5 |      |
|          | 选修 | 6    | 4、实践教学模块 | 27.5 |      |
| 2、学科基础平台 |    | 45.5 | 专业实践课程   | 必修   | 27.5 |
| 专业学科基础课程 | 必修 | 39.5 |          |      |      |
|          | 选修 | 6    | 5、素质拓展模块 |      | 6    |

\*通识教育选修课6学分包括：理工类、医学类、人文社科类、经济管理类中选择2学分（学生在本专业所属类别外的3个类中选择2个学分）；思想政治及新时代素质教育类选择2学分（“四史”1个学分，国家安全教育1个学分）；美育教育类选择2学分（美学和艺术史论类、艺术鉴赏和评论类课程至少选修1个学分）。

### VIII. Credits required for graduation: 160 credits

| Type of courses                |                  | Academic credits | Type of courses                    |                  | Academic credits |
|--------------------------------|------------------|------------------|------------------------------------|------------------|------------------|
| 1.Common Courses               |                  | 43               | 3. Specialized Courses             |                  | 38               |
| Common Basic Courses           |                  | 28               | Required Courses                   |                  | 27.5             |
| General Education Courses      | Required Courses | 9                | Elective Courses                   |                  | 10.5             |
|                                | Elective Courses | 6                | 4.Practicum and Internship Courses |                  | 27.5             |
| 2.General Disciplinary Courses |                  | 45.5             | Disciplinary Practical Courses     | Required Courses | 27.5             |
| Disciplinary Basic Courses     | Required Courses | 39.5             |                                    |                  |                  |
|                                | Elective Courses | 6                | 5.Quality Development Courses      |                  | 66               |

### 九、学分比例

#### IX.Ratio of Credits

##### 1、必修选修学分比例

The proportion of compulsory elective credits

| 类别 | 学分    | 占总学分比例 |
|----|-------|--------|
| 必修 | 137.5 | 85.9%  |
| 选修 | 22.5  | 14.1%  |

##### 2、实践教学环节学分比例

The Proportion of credits in practice teaching

| 类别     |        | 学分   | 占总学分比例 |
|--------|--------|------|--------|
| 实践教学环节 | 实验教学学分 | 21.7 | 34.9%  |
|        | 实践教学模块 | 27.5 |        |
|        | 素质拓展模块 | 6    |        |

### 十、辅修

修读本专业辅修课程达到26学分且主修专业达到毕业要求者，颁发辅修专业证书。修读本专业辅修课程达到40学分（含实践教学），并达到辅修学位授予条件的，颁发辅修学位证书。

#### X.Minor

The minor certificate will be issued to those who have completed 26 credits of minor courses and their own major has met the graduation requirements. The minor degree certificate will be issued to those who have completed 40 credits of minor courses (including practical teaching ) and meet the conditions for awarding minor degree.

十一、毕业要求实现矩阵

XI. Graduation Realization Matrix

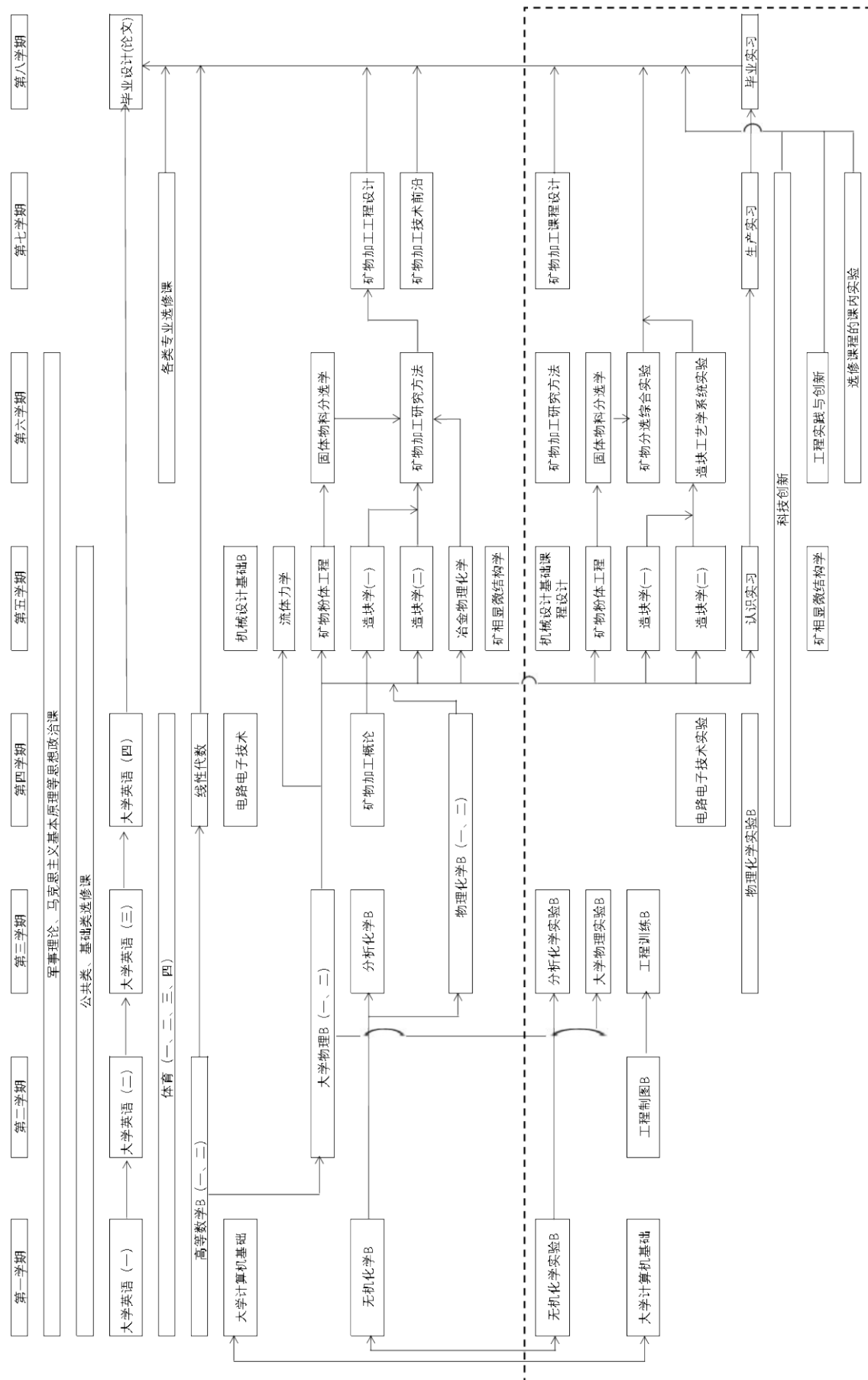
| 课程名称                 | 矿物加工工程专业毕业要求 |     |     |     |     |     |     |     |     |
|----------------------|--------------|-----|-----|-----|-----|-----|-----|-----|-----|
|                      | (1)          | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| 思想道德与法治              | √            |     |     |     |     | √   |     |     |     |
| 中国近现代史纲要             | √            |     |     |     |     |     |     |     |     |
| 马克思主义基本原理            | √            |     |     |     |     |     |     |     |     |
| 毛泽东思想和中国特色社会主义理论体系概论 | √            |     |     |     |     | √   |     |     |     |
| 习近平新时代中国特色社会主义思想概论   | √            |     |     |     |     | √   |     |     |     |
| 军事理论课                | √            |     |     |     |     |     |     |     |     |
| 大学生心理健康教育            | √            |     |     |     |     |     | √   |     | √   |
| 职业生涯规划与就业创业指导        | √            |     |     |     |     |     | √   |     | √   |
| 大学英语                 |              |     |     |     | √   |     | √   | √   | √   |
| 体育                   | √            |     |     |     |     |     | √   |     |     |
| 公益劳动                 | √            |     |     |     |     |     | √   |     |     |
| 形势与政策                | √            |     |     |     |     | √   |     | √   |     |
| 大学生创新创业基础            | √            |     |     |     |     |     |     |     | √   |
| 大学计算机基础              |              |     |     | √   | √   |     | √   |     | √   |
| C 语言程序设计基础           |              | √   |     | √   |     |     |     |     |     |
| C 语言程序设计基础实验         | √            | √   |     | √   |     |     |     |     |     |
| 人文社科类公选课             | √            |     |     |     |     | √   | √   | √   |     |
| 经济管理类公选课             |              | √   |     |     |     | √   | √   |     |     |
| 医学类公选课               |              | √   |     |     |     |     |     |     |     |
| 思想政治及新时代素质教育类公选课     | √            |     |     |     |     |     |     |     |     |
| 美育教育类公选课             | √            |     |     |     |     |     |     |     |     |
| 工程制图 B               |              | √   |     | √   | √   |     |     |     |     |
| 机械设计基础 B             |              | √   |     | √   | √   |     |     |     |     |
| 电路电子技术               |              | √   |     | √   |     |     |     |     |     |
| 电路电子技术实验             | √            | √   |     | √   |     |     | √   |     |     |
| 高等数学 B               |              | √   |     | √   | √   |     |     |     |     |
| 线性代数 B               |              | √   |     | √   | √   |     |     |     |     |
| 概率论与数理统计 B           |              | √   |     | √   | √   | √   |     |     |     |

| 课程名称        | 矿物加工工程专业毕业要求 |     |     |     |     |     |     |     |     |
|-------------|--------------|-----|-----|-----|-----|-----|-----|-----|-----|
|             | (1)          | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| 大学物理 B      |              | √   |     | √   |     |     |     |     |     |
| 大学物理实验 B    | √            | √   |     | √   |     |     | √   |     |     |
| 工程力学 B      |              | √   |     | √   |     |     |     |     |     |
| 无机化学 B      |              | √   |     | √   |     |     |     |     |     |
| 无机化学实验 B    | √            | √   |     | √   |     |     | √   |     |     |
| 有机化学 B      |              | √   |     | √   |     |     |     |     |     |
| 有机化学实验 B    | √            | √   |     | √   |     |     | √   |     |     |
| 物理化学 B      |              | √   |     | √   |     |     |     |     |     |
| 物理化学实验 B    | √            | √   |     | √   |     |     | √   |     |     |
| 分析化学 B      |              | √   |     | √   |     |     |     |     |     |
| 分析化学实验 B    | √            | √   |     | √   |     |     | √   |     |     |
| 冶金物理化学      |              | √   |     | √   |     |     |     |     |     |
| 流体力学        |              | √   |     | √   | √   |     |     |     |     |
| 矿物加工概论      |              |     | √   |     |     |     |     | √   |     |
| 矿物加工专业英语    |              |     |     |     |     |     | √   | √   | √   |
| 造块学         |              |     | √   | √   |     | √   |     | √   |     |
| 固体物料分选学     |              |     | √   | √   |     | √   |     | √   |     |
| 矿物加工研究方法    |              |     | √   | √   | √   | √   |     | √   | √   |
| 矿物加工工程设计    |              |     | √   | √   | √   | √   | √   | √   |     |
| 造块工艺学系统实验   | √            |     | √   | √   |     |     | √   |     | √   |
| 矿物分选综合实验    | √            |     | √   | √   |     |     | √   |     | √   |
| 矿物加工技术前沿    |              |     |     |     |     |     |     | √   |     |
| 矿物材料        |              |     | √   | √   |     | √   |     | √   |     |
| 技术经济与投资分析   |              | √   |     | √   |     | √   | √   | √   |     |
| 矿山企业管理      |              | √   |     | √   |     | √   | √   |     |     |
| 二次资源综合利用    | √            |     | √   | √   |     | √   |     | √   |     |
| 矿物显微结构学     |              | √   | √   | √   | √   |     |     |     |     |
| 炼铁学         |              |     | √   | √   |     | √   |     | √   |     |
| 资源分离与提取技术原理 |              |     | √   | √   |     | √   |     | √   | √   |
| 固体废物资源化     | √            |     | √   | √   |     | √   |     | √   | √   |
| 废水资源化综合利用   | √            |     | √   | √   |     | √   |     | √   | √   |

| 课程名称         | 矿物加工工程专业毕业要求 |     |     |     |     |     |     |     |     |
|--------------|--------------|-----|-----|-----|-----|-----|-----|-----|-----|
|              | (1)          | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| 再生资源工艺设备     |              |     | √   | √   |     |     |     | √   | √   |
| 再生资源工程设计     |              |     | √   | √   | √   | √   | √   | √   | √   |
| 计算机在矿物加工中的应用 |              |     |     | √   | √   |     | √   | √   |     |
| 矿物加工过程检测与控制  |              |     |     | √   | √   |     |     | √   |     |
| 矿物化学提取       |              |     | √   | √   |     | √   |     | √   |     |
| CAD 技术       |              |     |     | √   | √   |     | √   |     |     |
| 环境影响评价       | √            | √   |     |     |     | √   |     |     |     |
| 冶金炉原理        |              | √   |     | √   |     |     |     |     |     |
| 采矿概论         |              |     |     |     |     | √   |     | √   |     |
| 选矿药剂         |              |     | √   | √   | √   | √   |     | √   |     |
| 矿物加工工程典型案例分析 | √            |     | √   | √   | √   | √   |     |     |     |
| 信息检索与利用      |              |     |     |     | √   |     |     |     |     |
| 军事训练         | √            |     |     |     |     |     | √   |     |     |
| 工程训练 B       | √            |     |     | √   |     |     | √   |     |     |
| 机械设计基础课程设计   | √            | √   |     | √   | √   |     | √   |     |     |
| 认识实习         | √            | √   | √   | √   |     |     | √   | √   | √   |
| 生产实习         | √            | √   | √   | √   |     | √   | √   | √   | √   |
| 科技创新         | √            | √   | √   | √   | √   | √   | √   | √   | √   |
| 工程实践与创新      | √            | √   | √   | √   |     | √   | √   | √   | √   |
| 矿物加工课程设计     | √            | √   | √   | √   |     |     | √   |     | √   |
| 毕业实习         | √            | √   | √   | √   |     | √   | √   | √   | √   |
| 毕业设计（论文）     | √            | √   | √   | √   | √   | √   | √   | √   | √   |
| 创新创业教育       | √            |     |     |     |     |     |     |     | √   |
| 第二课堂         | √            |     |     |     |     |     | √   |     | √   |

## 十二、课程修读进程表

## XII. Course review Process Map



十三、教学环节设置及学分分布表 (四年制本科 2024级 矿物加工工程专业“卓越计划”)

XIII. Offered Course and Distribution of Academic Credits

| 课 程 类 型 |          |        | 课程性质 | 课 程 编 码 | 课 程 名 称                                                                                                                 | 学 分 | 合 计 | 课 内 学 时 |     |     | 实践学时 | 学 期 | 是否辅修/双学位课程 | 先 修 课 程/备 注 |
|---------|----------|--------|------|---------|-------------------------------------------------------------------------------------------------------------------------|-----|-----|---------|-----|-----|------|-----|------------|-------------|
|         |          |        |      |         |                                                                                                                         |     |     | 讲 课     | 实 验 | 上 机 |      |     |            |             |
| 平台      | 公共课程平台课程 | 公共基础课程 | 必修   | 5105017 | 思想道德与法制<br>Ideology and morality and rule of law                                                                        | 3   | 48  | 42      | 0   | 0   | 6    | 1   |            |             |
|         |          |        |      | 5103001 | 中国近现代史纲要<br>An Outline of Modern and Contemporary History of China                                                      | 3   | 48  | 42      | 0   | 0   | 6    | 2   |            |             |
|         |          |        |      | 5102001 | 马克思主义基本原理<br>Fundamentals of Marxism                                                                                    | 3   | 48  | 44      | 0   | 0   | 4    | 3   |            |             |
|         |          |        |      | 5101005 | 毛泽东思想与中国特色社会主义理论体系概论<br>Theoretical system of socialism with Chinese characteristics                                    | 3   | 48  | 32      | 0   | 0   | 16   | 4   |            |             |
|         |          |        |      | 5101006 | 习近平新时代中国特色社会主义思想理论体系概论<br>An Introduction of Xi Jinping Through on Socialism with Chinese Characteristics for a New Era | 3   | 48  | 32      | 0   | 0   | 16   | 4   |            |             |
|         |          |        |      | 1401048 | 大学综合英语(一)<br>College English (I)                                                                                        | 2   | 32  | 32      | 0   | 0   | 0    | 1   |            |             |
|         |          |        |      | 1401072 | 大学综合英语(二)<br>College English (II)                                                                                       | 2   | 32  | 32      | 0   | 0   | 0    | 2   |            |             |
|         |          |        |      | 1401049 | 大学综合英语(三)<br>College English (III)                                                                                      | 2   | 32  | 32      | 0   | 0   | 0    | 3   |            |             |
|         |          |        |      | 1401074 | 大学综合英语(四)<br>College English (IV)                                                                                       | 1   | 16  | 16      | 0   | 0   | 0    | 1   |            |             |
|         |          |        |      | 1501882 | 体育(一)<br>Physical Education(I)                                                                                          | 1   | 26  | 26      | 0   | 0   | 0    | 1   |            |             |
|         |          |        |      | 1501883 | 体育(二)<br>Physical Education(II)                                                                                         | 1   | 34  | 34      | 0   | 0   | 0    | 2   |            |             |
|         |          |        |      | 1501884 | 体育(三)<br>Physical Education(III)                                                                                        | 1   | 34  | 34      | 0   | 0   | 0    | 3   |            |             |
|         |          |        |      | 1501885 | 体育(四)<br>Physical Education(IV)                                                                                         | 1   | 34  | 34      | 0   | 0   | 0    | 4   |            |             |
|         |          |        |      | 5106001 | 形势与政策<br>World Affairs and State Policy                                                                                 | 2   | 64  | 64      | 0   | 0   | 0    | 1-8 |            | 分散进行        |
|         |          |        |      | 1501024 | 体质健康测试<br>Physical Health Test                                                                                          | 0   | 16  | 0       | 0   | 0   | 16   | 1-8 |            | 分散进行        |
|         | 通识教育课程   |        | 必修   | 1306009 | 大学计算机基础<br>Computer Foundation                                                                                          | 2   | 32  | 20      | 0   | 12  | 0    | 1   |            |             |

教学环节设置及学分分布表 (四年制本科 2024级 矿物加工工程专业“卓越计划”)

Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质 | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称                                                           | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------|------------------|----------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |          |                  |                                                                            |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 平台               | 公共课程平台课程 | 必修               | 2501016 大学生劳动教育<br>Labor Education for College Students                    | 1      | 32     | 2       | 0   | 0   | 30               | 1-4    |                        | 分散进行           |
|                  |          |                  | 2502006 大学生心理健康教育<br>Mental Health Education                               | 2      | 32     | 32      | 0   | 0   | 0                | 1      |                        |                |
|                  |          |                  | 2503001 职业生涯规划与就业创业指导<br>Career Plan and Vocational Guidance               | 1      | 16     | 16      | 0   | 0   | 0                | 2      |                        |                |
|                  |          |                  | 2504005 军事理论<br>Military Theory                                            | 2      | 36     | 36      | 0   | 0   | 0                | 2      |                        |                |
|                  |          |                  | 8001003 大学生创新创业基础<br>Fundamentals of entrepreneurship                      | 1      | 16     | 16      | 0   | 0   | 0                | 2      |                        |                |
|                  |          | 选修               | 人文社科类1学分<br>Humanity and Social Science 1 Academic Credits                 |        |        |         |     |     |                  |        |                        |                |
|                  |          |                  | 经济管理类1学分<br>Economic and Management 1 Academic Credits                     |        |        |         |     |     |                  |        |                        |                |
|                  |          |                  | 医学类1学分<br>Medicine 1 Academic Credits                                      |        |        |         |     |     |                  |        |                        |                |
|                  |          |                  | 思想政治及新时代素质教育类2学分<br>Ideological and Political Education 2 Academic Credits |        |        |         |     |     |                  |        |                        |                |
|                  |          |                  | 美育教育类2学分<br>Art Education 2 Academic Credits                               |        |        |         |     |     |                  |        |                        |                |
|                  |          |                  |                                                                            |        |        |         |     |     |                  |        |                        |                |
|                  |          | 必修               | 0702603 高等数学B(一)<br>Advanced Mathematics B (I)                             | 4      | 64     | 64      | 0   | 0   | 0                | 1      |                        |                |
|                  |          |                  | 0702604 高等数学B(二)<br>Advanced Mathematics B (II)                            | 5      | 80     | 80      | 0   | 0   | 0                | 2      |                        |                |
|                  |          |                  | 2206665 无机化学B<br>Inorganic Chemistry B                                     | 3      | 48     | 48      | 0   | 0   | 0                | 1      |                        |                |
|                  |          |                  | 2253020 无机化学实验B<br>Experiments in Inorganic Chemistry B                    | 1      | 16     | 0       | 16  | 0   | 0                | 1      |                        |                |
|                  |          |                  | 0302609 工程制图B<br>Engineering Drawing B                                     | 3      | 48     | 40      | 0   | 8   | 0                | 1      |                        |                |
|                  |          |                  | 0703605 大学物理B(一)<br>College Physics B(I)                                   | 2.5    | 40     | 40      | 0   | 0   | 0                | 2      |                        |                |
|                  |          |                  | 0703606 大学物理B (二)<br>College Physics B(II)                                 | 2      | 32     | 32      | 0   | 0   | 0                | 3      |                        |                |

教学环节设置及学分分布表 (四年制本科 2024级 矿物加工工程专业“卓越计划”)  
Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质 | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称                                                                   | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------|------------------|------------------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |          |                  |                                                                                    |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 平台               | 学科基础平台   | 专业学科基础课程         | 0703607<br>大学物理实验B<br>Experiments in College Physics B                             | 1      | 24     | 0       | 24  | 0   | 0                | 3      |                        |                |
|                  |          |                  | 2206681<br>分析化学B<br>Analytical Chemistry B                                         | 2      | 32     | 32      | 0   | 0   | 0                | 3      |                        |                |
|                  |          |                  | 2253051<br>分析化学实验B<br>Experiments in Analytical Chemistry B                        | 1      | 24     | 0       | 24  | 0   | 0                | 3      |                        |                |
|                  |          |                  | 0401039<br>电路电子技术<br>Electric Circuit and Electronic Technologies                  | 3      | 48     | 48      | 0   | 0   | 0                | 4      |                        |                |
|                  |          |                  | 0450005<br>电路电子技术实验<br>Experiments of Electric Circuit and Electronic Technologies | 1      | 16     | 0       | 16  | 0   | 0                | 4      |                        |                |
|                  |          |                  | 2206675<br>物理化学B(一)<br>Physical Chemistry B (I)                                    | 2      | 32     | 32      | 0   | 0   | 0                | 3      |                        |                |
|                  |          |                  | 2206676<br>物理化学B(二)<br>Physical Chemistry B (II)                                   | 1.5    | 24     | 24      | 0   | 0   | 0                | 4      |                        |                |
|                  |          |                  | 2253054<br>物理化学实验B(一)<br>Experiments in Physical Chemistry B (I)                   | 1      | 16     | 0       | 16  | 0   | 0                | 3      |                        |                |
|                  |          |                  | 2253022<br>物理化学实验B(二)<br>Experiments in Physical Chemistry B (I)                   | 1      | 16     | 0       | 16  | 0   | 0                | 4      |                        |                |
|                  |          |                  | 0702104<br>线性代数B<br>Linear Algebra B                                               | 2      | 32     | 32      | 0   | 0   | 0                | 4      |                        |                |
|                  |          |                  | 0304602<br>机械设计基础B<br>Basics of Mechanical Design B                                | 3.5    | 56     | 50      | 6   | 0   | 0                | 5      |                        |                |
|                  |          | 必修               | 0702304<br>概率论与数理统计B<br>Probability Theory and Mathematical Statistics B           | 2.5    | 40     | 40      | 0   | 0   | 0                | 3      |                        |                |
|                  |          |                  | 2206679<br>有机化学B<br>Organic Chemistry B                                            | 2.5    | 40     | 40      | 0   | 0   | 0                | 3      |                        | 同时<br>选修       |
|                  |          |                  | 2253027<br>有机化学实验B<br>Experiments in Organic Chemistry                             | 1      | 24     | 0       | 24  | 0   | 0                | 3      |                        |                |
|                  |          |                  | 1306010<br>C语言程序设计基础<br>Basics of Computer Programming (C)                         | 2.5    | 40     | 40      | 0   | 0   | 0                | 3      |                        | 同时<br>选修       |
|                  |          |                  | 1306011<br>C语言程序设计基础实验<br>Experiments in Computer Programming (C)                  | 1      | 24     | 0       | 0   | 0   | 24               | 3      |                        |                |

教学环节设置及学分分布表 (四年制本科 2024级 矿物加工工程专业“卓越计划”)

Offered Course and Distribution of Academic Credits

| 课 程 类 型 |        | 课程性质   | 课 程 编 码 | 课 程 名 称 | 学 分            | 合 计 | 课 内 学 时 |    |    | 实 践 学 时 | 学 期 | 是否辅修/双学位课程 | 先 修 课 程/备 注 |
|---------|--------|--------|---------|---------|----------------|-----|---------|----|----|---------|-----|------------|-------------|
| 平台      | 学科基础平台 | 专业基础课程 | 必修      | 0701606 | 工程力学B          | 3   | 48      | 42 | 6  | 0       | 0   | 4          |             |
| 模块      |        | 专业必修课程 | 必修      | 0107046 | 矿物加工概论         | 2   | 32      | 32 | 0  | 0       | 0   | 4          | 是           |
|         |        |        |         | 0107079 | 矿物加工技术前沿       | 1.5 | 24      | 24 | 0  | 0       | 0   | 7          |             |
|         |        |        |         | 0107110 | 造块学（一）：烧结理论与工艺 | 2.5 | 40      | 36 | 4  | 0       | 0   | 5          | 是           |
|         |        |        |         | 0107111 | 造块学(二)：团矿理论与工艺 | 2.5 | 40      | 36 | 4  | 0       | 0   | 5          | 是           |
|         |        |        |         | 0107009 | 流体力学           | 2   | 32      | 32 | 0  | 0       | 0   | 5          | 是           |
|         |        |        |         | 0107065 | 冶金物理化学         | 2   | 32      | 32 | 0  | 0       | 0   | 5          | 是           |
|         |        |        |         | 0107115 | 矿物粉体工程         | 2   | 32      | 28 | 4  | 0       | 0   | 5          | 是           |
|         |        |        |         | 0107116 | 固体物料分选学        | 3.5 | 56      | 50 | 6  | 0       | 0   | 6          | 是           |
|         |        |        |         | 0107054 | 矿物加工研究方法       | 3   | 48      | 42 | 6  | 0       | 0   | 6          | 是           |
|         |        |        |         | 0107067 | 造块工艺学系统实验      | 1.5 | 24      | 0  | 24 | 0       | 0   | 6          | 是           |
|         |        |        |         | 0107066 | 矿物分选综合实验       | 1.5 | 24      | 0  | 24 | 0       | 0   | 6          | 是           |
|         |        |        |         | 0107053 | 矿物加工工程设计       | 3.5 | 56      | 48 | 0  | 8       | 0   | 7          | 是           |
|         |        | 专业选修课程 | 选修      | 0107086 | CAD技术          | 2   | 32      | 18 | 0  | 14      | 0   | 6          |             |
|         |        |        |         | 0107084 | 中国矿冶科技史        | 2   | 32      | 32 | 0  | 0       | 0   | 4          |             |

教学环节设置及学分分布表

(四年制本科 2024级 矿物加工工程专业“卓越计划”)

Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质       | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称 | 学<br>分      | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------------|------------------|------------------|-------------|--------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |                |                  |                  |             |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 模块               | 专业<br>课程<br>模块 | 专业<br>选修<br>课程   | 1601004          | 信息检索与利用     | 1      | 16      | 12  | 0   | 4                | 0      | 4                      |                |
|                  |                |                  | 0107068          | 资源分离与提取技术原理 | 2.5    | 40      | 40  | 0   | 0                | 0      | 4                      |                |
|                  |                |                  | 0101194          | 矿物显微结构学     | 2.5    | 40      | 30  | 10  | 0                | 0      | 5                      |                |
|                  |                |                  | 0107012          | 固体废物资源化     | 2      | 32      | 32  | 0   | 0                | 0      | 5                      |                |
|                  |                |                  | 0107070          | 废水资源化综合利用   | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 0107071          | 再生资源工艺设备    | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 0107035          | 炼铁学         | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 0107050          | 矿物化学提取      | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 0107102          | 矿物加工专业英语    | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 0107073          | 矿山企业管理      | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 0101223          | 技术经济与投资分析   | 1.5    | 24?     | 24? | 0?  | 0?               | 0?     | 6                      |                |
|                  |                |                  | 0101152          | 采矿概论        | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 0107078          | 矿物加工过程检测与控制 | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 0107094          | 城市矿产与循环经济   | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 0107104          | 3D设计及打印技术   | 1.5    | 24      | 14  | 0   | 10               | 0      | 7                      |                |
|                  |                |                  | 0107056          | 二次资源综合利用    | 2      | 32      | 32  | 0   | 0                | 0      | 7                      |                |

教学环节设置及学分分布表 (四年制本科 2024级 矿物加工工程专业“卓越计划”)

Offered Course and Distribution of Academic Credits

| 课 程 类 型 |        |        | 课程性质 | 课 程 编 码 | 课 程 名 称                                                                  | 学 分 | 合 计   | 课 内 学 时 |     |     | 实践学时  | 学 期        | 是否辅修/双学位课程 | 先 修 课 程/备 注 |
|---------|--------|--------|------|---------|--------------------------------------------------------------------------|-----|-------|---------|-----|-----|-------|------------|------------|-------------|
|         |        |        |      |         |                                                                          |     |       | 讲 课     | 实 验 | 上 机 |       |            |            |             |
| 模块      | 专业基础课程 | 必修     | 必修   | 0106099 | 环境影响评价<br>Environmental Assessment                                       | 2   | 32    | 32      | 0   | 0   | 0     | 7          |            |             |
|         |        |        |      | 0107103 | 计算机在矿物加工中的应用<br>Computer Application of Mineral Processing               | 2   | 32    | 28      | 0   | 4   | 0     | 7          |            |             |
|         |        |        |      | 0107052 | 矿物材料<br>Mineral Material                                                 | 2   | 32    | 32      | 0   | 0   | 0     | 7          |            |             |
|         |        |        |      | 0107072 | 再生资源工程设计<br>Design of Renewable Resources                                | 2   | 32    | 32      | 0   | 0   | 0     | 7          |            |             |
|         |        |        |      | 0107074 | 选矿药剂<br>Mineral Processing Reagents                                      | 2   | 32    | 32      | 0   | 0   | 0     | 7          |            |             |
|         |        |        |      | 0107075 | 再生资源生物技术<br>Biological Technology of Renewable Resources                 | 2   | 32    | 32      | 0   | 0   | 0     | 7          |            |             |
|         |        |        |      | 0107108 | 矿物加工分析测试技术<br>Mineral processing analysis and testing technology         | 2   | 32    | 32      | 0   | 0   | 0     | 7          |            |             |
|         |        |        |      | 0107107 | 矿业环境工程<br>Mining Environmental Engineering                               | 2   | 32    | 32      | 0   | 0   | 0     | 7          |            |             |
|         |        |        |      | 0107087 | 矿物加工工程典型案例分析<br>Typical Cases Analysis of Mineral Processing Engineering | 1.5 | 24    | 24      | 0   | 0   | 0     | 7          |            |             |
|         | 实践教学模块 | 专业实践课程 | 必修   | 2504006 | 军事训练<br>Military Training                                                | 2   | 112   | 0       | 0   | 0   | 112   | 1          |            |             |
|         |        |        |      | 1701008 | 工程训练B<br>Engineering Training B                                          | 1.5 | 3 (周) | 0       | 0   | 0   | 3 (周) | 3          |            |             |
|         |        |        |      | 0107091 | 科技创新<br>Technology Innovation                                            | 2   | 2 (周) | 0       | 0   | 0   | 2 (周) | 4, 5, 6, 7 |            |             |
|         |        |        |      | 0107030 | 认识实习<br>Introductory Practice                                            | 2   | 2 (周) | 0       | 0   | 0   | 2 (周) | 5          |            |             |
|         |        |        |      | 0304010 | 机械设计基础课程设计<br>Course Project in Basics of Mechanical Design              | 1   | 2 (周) | 0       | 0   | 0   | 2 (周) | 5          |            |             |
|         |        |        |      | 0107031 | 生产实习<br>Production Practice                                              | 3   | 3 (周) | 0       | 0   | 0   | 3 (周) | 7          | 是          |             |

教学环节设置及学分分布表

(四年制本科 2024级 矿物加工工程专业“卓越计划”)

Offered Course and Distribution of Academic Credits

| 课 程 类 型 |        |        | 课程性质 | 课 程 编 码                                              | 课 程 名 称                                          | 学 分 | 合 计       | 课 内 学 时 |     |     | 实践学时      | 学 期 | 是否辅修/双学位课程 | 先 修 课 程/备 注 |
|---------|--------|--------|------|------------------------------------------------------|--------------------------------------------------|-----|-----------|---------|-----|-----|-----------|-----|------------|-------------|
|         |        |        |      |                                                      |                                                  |     |           | 讲 课     | 实 验 | 上 机 |           |     |            |             |
| 模块      | 实践教学模块 | 专业实践课程 | 必修   | 0107063                                              | 工程实践与创新<br>Engineering Training and Innovations  | 4   | 4<br>(周)  | 0       | 0   | 0   | 4<br>(周)  | 7   |            |             |
|         |        |        |      | 0107061                                              | 矿物加工课程设计<br>Course Project in Mineral Processing | 1   | 2<br>(周)  | 0       | 0   | 0   | 2<br>(周)  | 7   |            |             |
|         |        |        |      | 0107097                                              | 毕业实习<br>Pre-graduation Practice Experience       | 3   | 3<br>(周)  | 0       | 0   | 0   | 3<br>(周)  | 8   | 是          |             |
|         |        |        |      | 0107096                                              | 毕业设计(论文)<br>Graduation Project (Thesis)          | 8   | 14<br>(周) | 0       | 0   | 0   | 14<br>(周) | 8   | 是          |             |
|         | 素质拓展模块 | 创新创业教育 | 必修   | 创新创业实践3学分<br>Innovation Practices 3 Academic Credits |                                                  |     |           |         |     |     |           |     |            |             |
|         |        | 第二课堂   | 必修   | 第二课堂3学分<br>Second Classroom 3 Academic Credits       |                                                  |     |           |         |     |     |           |     |            |             |

\*创新创业教育和第二课堂主要安排在小学期进行